

Leimarembi Foundation

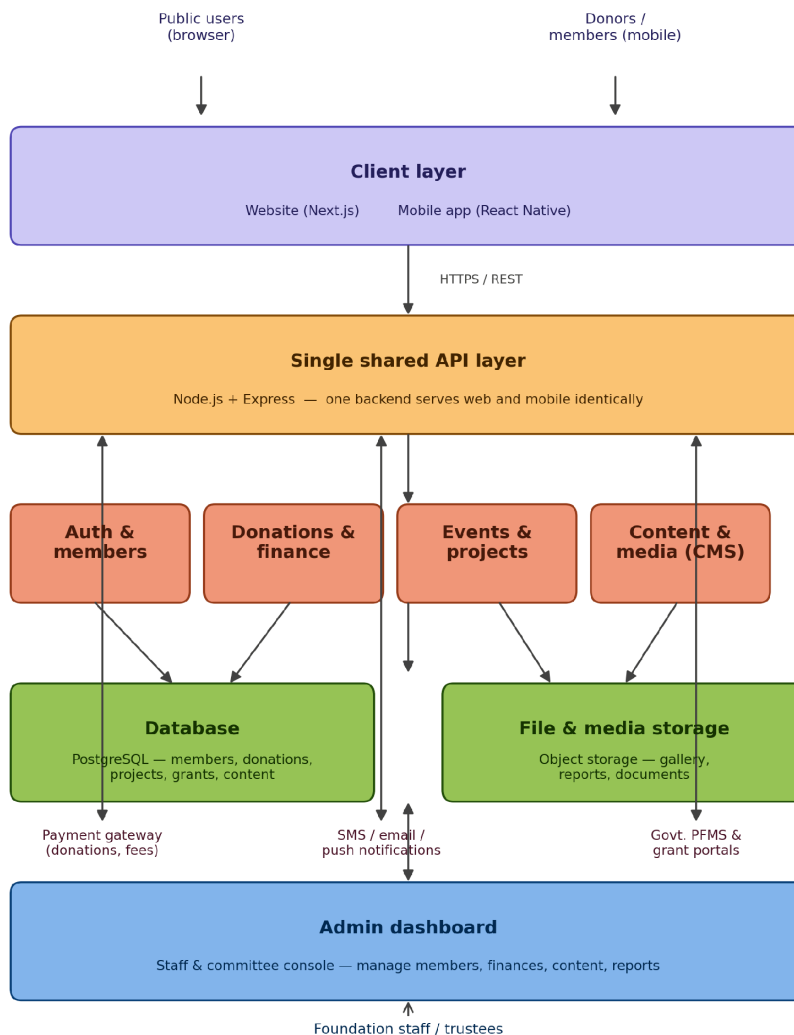
Digital Governance & Community Development Platform

Technical Architecture — Website & Mobile Application

1. Overview

This document sets out the technical architecture for the Foundation's website and mobile application. The design uses one shared backend for both the website and the mobile app, so every feature — membership, donations, events, content — is built once and works identically on both platforms. This keeps development faster now and maintenance simpler for years to come.

2. Architecture Diagram



3. Technology Stack

The stack below is selected for reliability, low ongoing cost, and ease of finding developers in India to maintain it long-term. Every layer uses widely-adopted, well-documented, actively-maintained technology.

Layer	Technology	Why this choice
Website frontend	Next.js (React)	Fast page loads, good SEO for public content, one codebase shared with reusable UI logic
Mobile app	React Native	Shares components and API logic with the website team; single JavaScript codebase builds both Android and iOS
Backend API	Node.js + Express	One API serves the website, mobile app, and admin dashboard identically — no duplicate logic
Database	PostgreSQL	Reliable, free, handles relational data (members, donations, projects) and scales with the Foundation
File & media storage	Object storage (S3-compatible)	Keeps large files (photos, reports, videos) out of the database, cheap and fast to serve
Hosting	Cloud VPS/PaaS with managed backups	Predictable cost, easy to scale, automatic daily backups
Admin dashboard	Same React codebase, role-restricted	Staff and trustees manage everything without touching code

4. How the Website and Mobile App Share One System

The website and the mobile app are two separate front-ends that both talk to the same backend API. A member who registers on the website automatically has an account in the mobile app, and vice versa. This is the standard approach used by every modern organisation running both a website and an app — it avoids double data entry, double bugs, and double maintenance cost.

- One database — member, donation, and event records exist in exactly one place.
- One API — business rules (who can donate, how membership renews) are written once.
- Two thin clients — the website and the app only handle display and input; they contain no duplicated logic.
- A change to a rule (e.g. new membership fee) is made once on the backend and takes effect everywhere immediately.

5. Core Modules

Mapped from the Foundation's requirements to concrete system components:

- Member management — registration, digital membership card, family & senior citizen records, status tracking
- Financial management — donations, fees, receipts, income & expenditure, audit reports, bank reconciliation
- Project & activity management — planning, monitoring, beneficiary records, progress reports
- Meeting management — notices, agendas, attendance, minutes, resolutions
- Government grant tracking — scheme database, applications, PFMS integration, utilization certificates
- Cultural preservation archive — heritage documents, recordings, songs and dance documentation
- Health & welfare records — check-ups, emergency contacts, medical camp management
- Digital library — trust deed, bye-laws, circulars, notifications, training materials

6. Security & Data Protection

- All traffic encrypted with HTTPS/TLS, on both website and mobile app
- Role-based access: members, staff, trustees, and administrators each see only what they're authorised to
- Passwords hashed, never stored in plain text; sessions use industry-standard tokens (JWT)
- Automated daily database backups, retained for 30 days, stored separately from the live server
- Financial and medical records restricted to authorised roles only, with an access log

7. Hosting & Deployment

The system is deployed on a managed cloud host with automatic scaling, SSL certificates, and daily backups. Staging and production environments are kept separate, so new features are tested before they reach the public site. This keeps monthly hosting cost predictable and low while the Foundation is small, with room to scale as membership grows.

8. Maintainability

The architecture is built so the Foundation is never dependent on one individual developer:

- Standard, widely-taught technologies (React, Node.js, PostgreSQL) — easy to find developers for in India
- Admin dashboard lets non-technical staff publish news, upload reports, and manage members without a developer
- Source code kept in a version-controlled repository (Git) with full history of every change
- Written technical documentation and a deployment runbook handed over with the finished system
- Automated backups and monitoring alert the team before small issues become outages

9. Suggested Rollout Plan

Phase	Deliverable	Duration
Phase 1	Public website: About, Activities, News, Gallery, Annual Reports, Contact, Donation page	3–4 weeks
Phase 2	Membership registration + admin dashboard (member database, content publishing)	2–3 weeks
Phase 3	Mobile app (Android & iOS) reusing the same backend API — digital membership card, notifications, directory	5–6 weeks
Phase 4	Financial management, meeting management, payment gateway integration	3 weeks
Phase 5	Government grant module, cultural preservation archive, health & welfare module	Ongoing, as funded
Phase 6	Testing and Deployment	3-4 weeks